

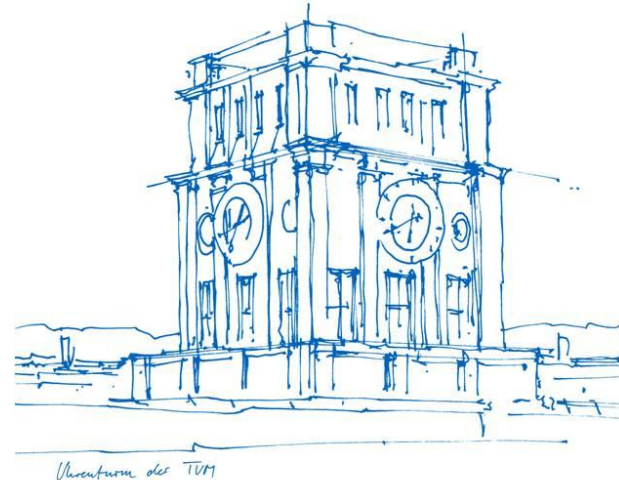
IFAN: An Explainability-Focused Interaction Framework for Humans and NLP Models

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Maximilian Kummeth, Kirill Gringauz,
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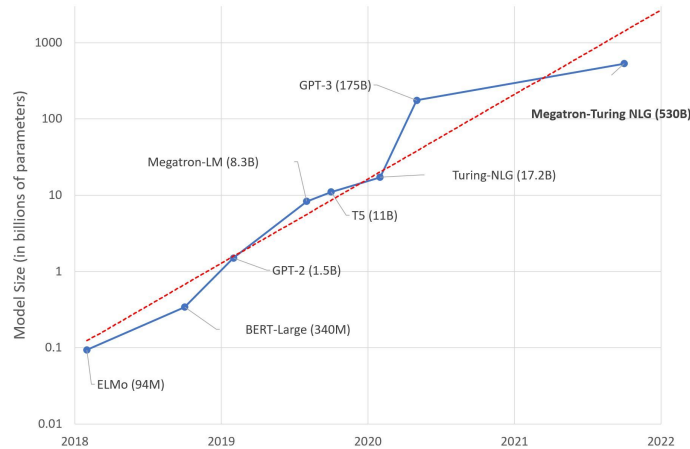
Technical University of Munich, Germany

AACL 2023

1st-4th October | Bali



The Current Stand



Performance

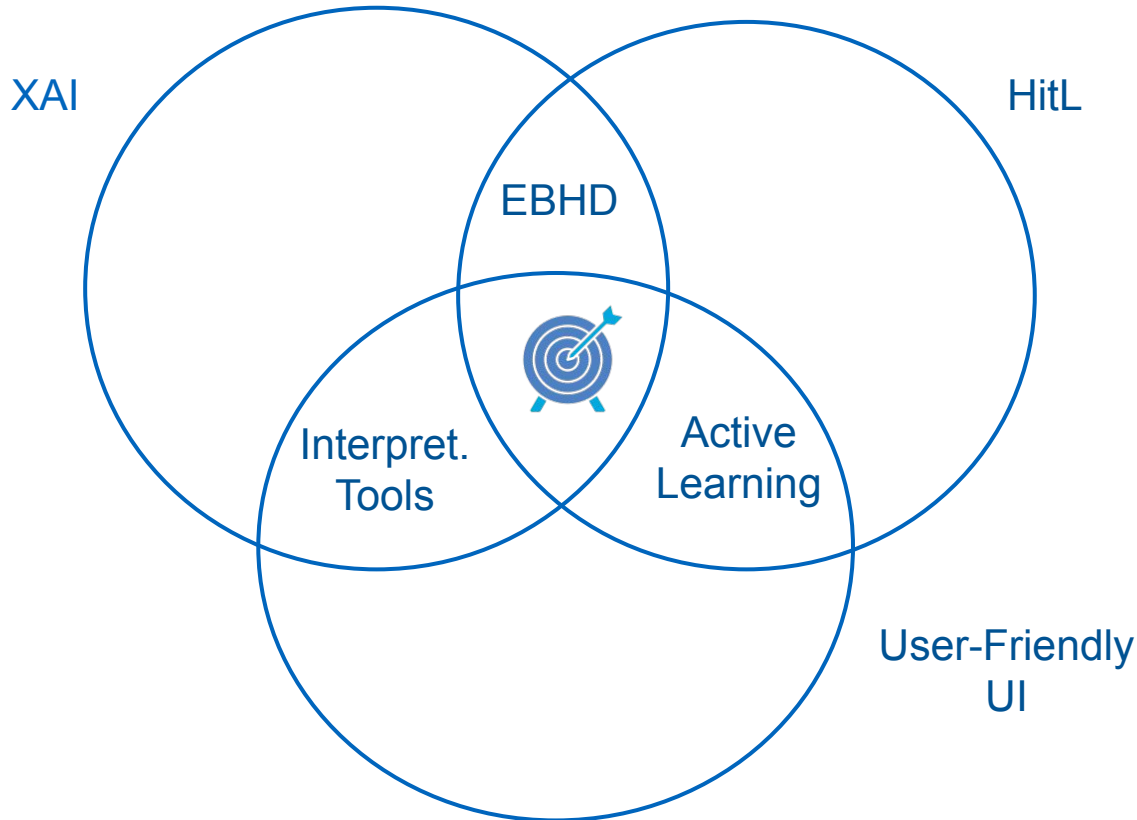


Interpretability

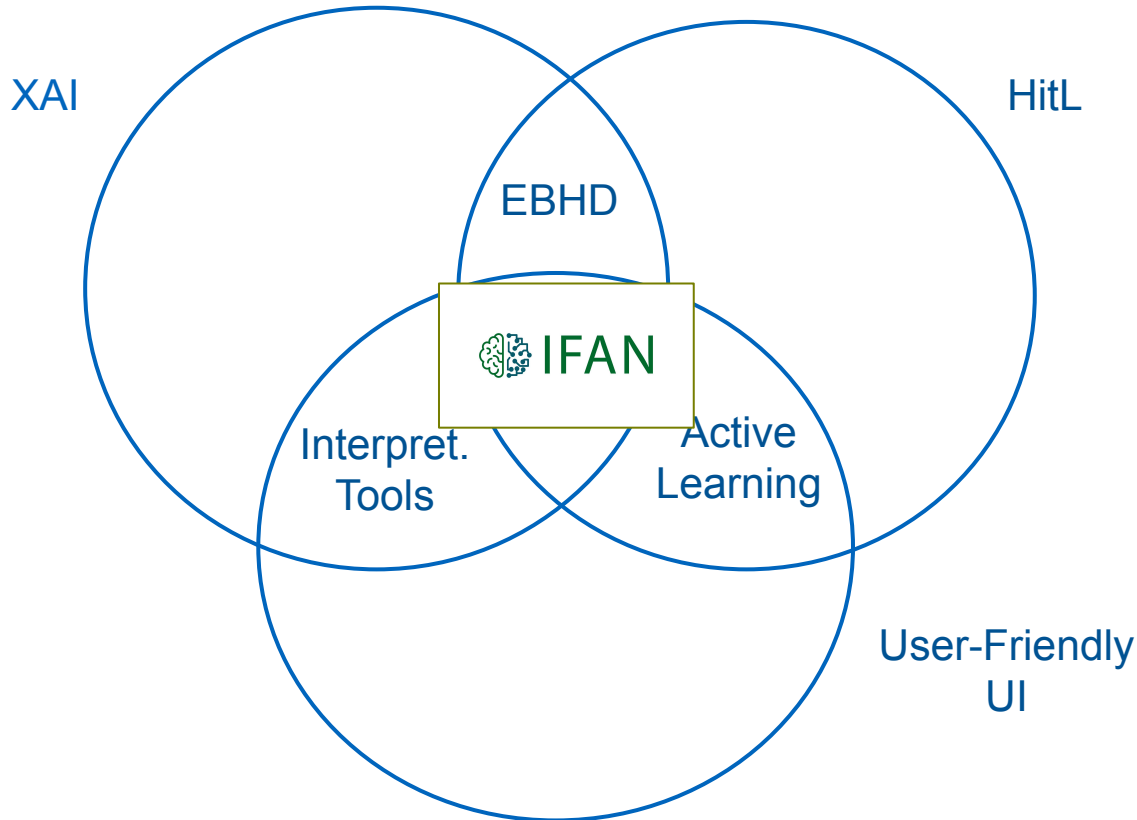


Controllability

The Current Stand



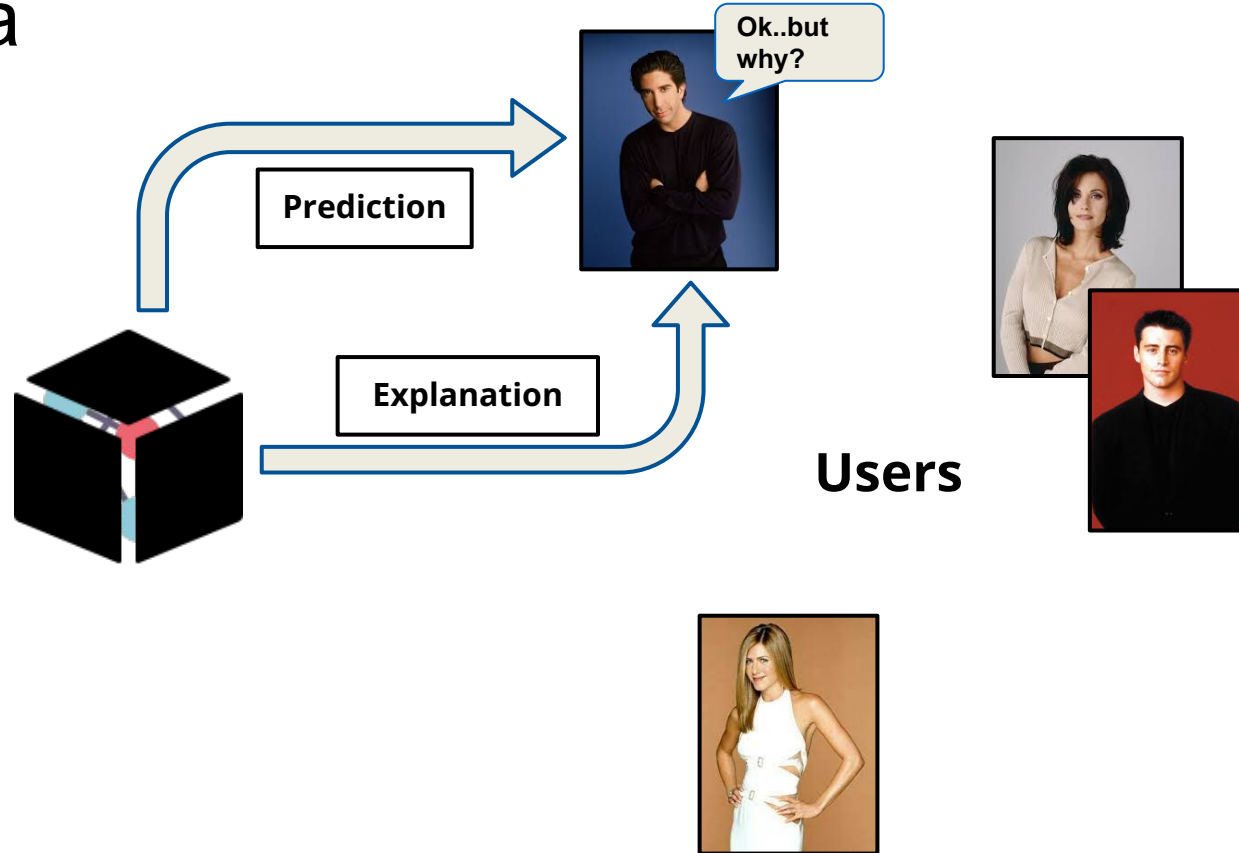
The Current Stand



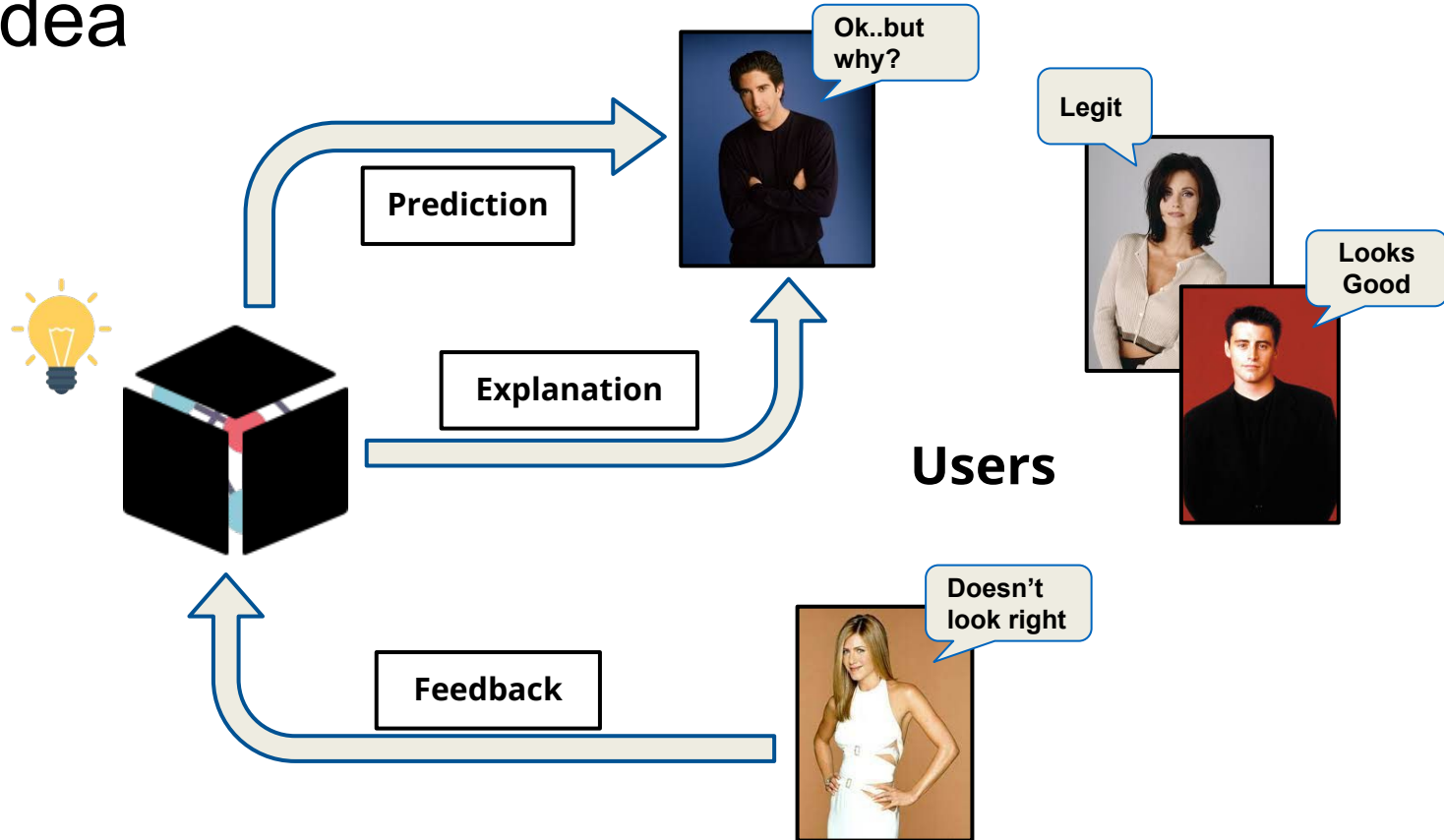
Our contribution

- 1 Propose IFAN: an EBDH Framework for NLP Models**
Users can observe explanations, edit the rationale, give feedback, etc.
- 2 Extend it with a UI and a Management Systems**
Monitor improvement, configure models and user access, etc.
- 3 Test IFAN on a Model Debiasing Task**
We propose feedback rebalancing to contrast model forgetfulness.

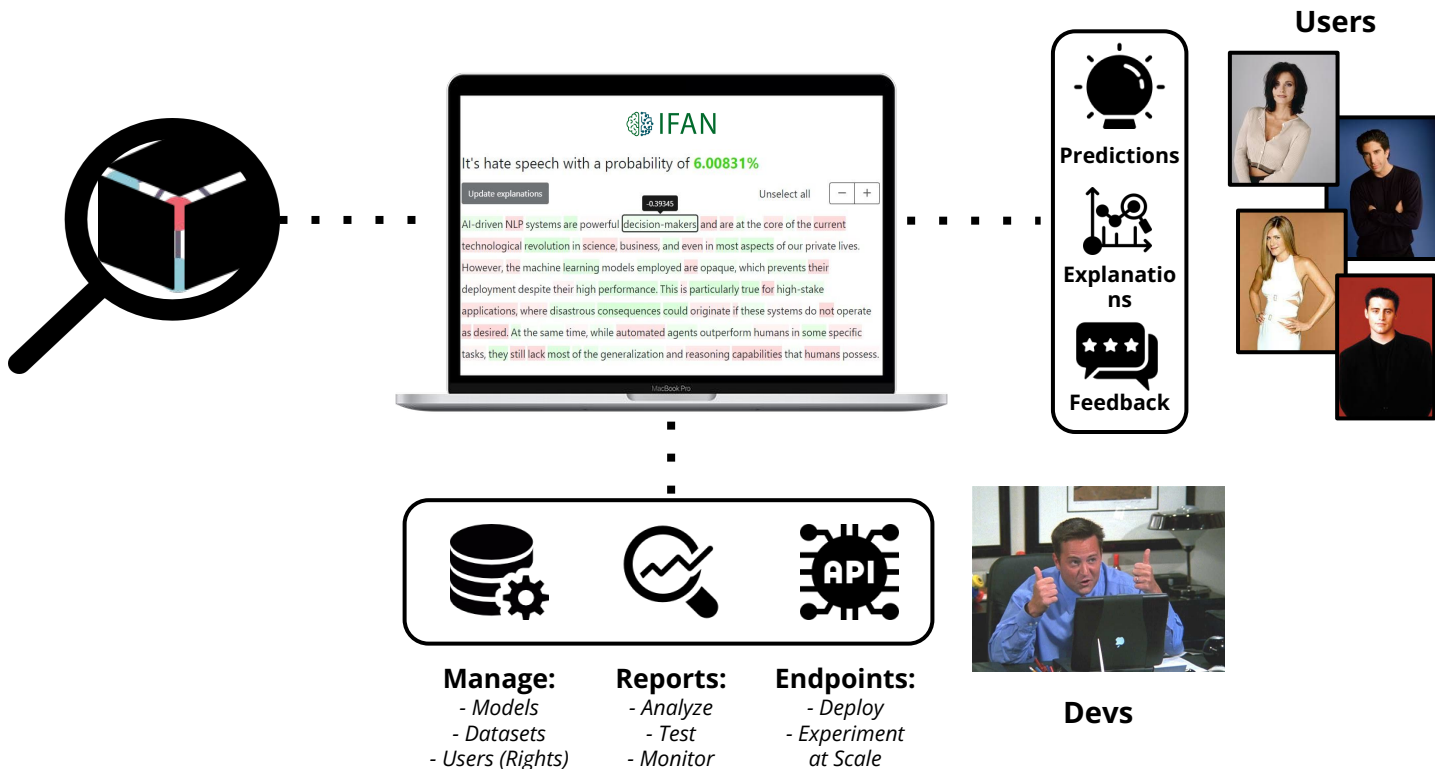
The Idea



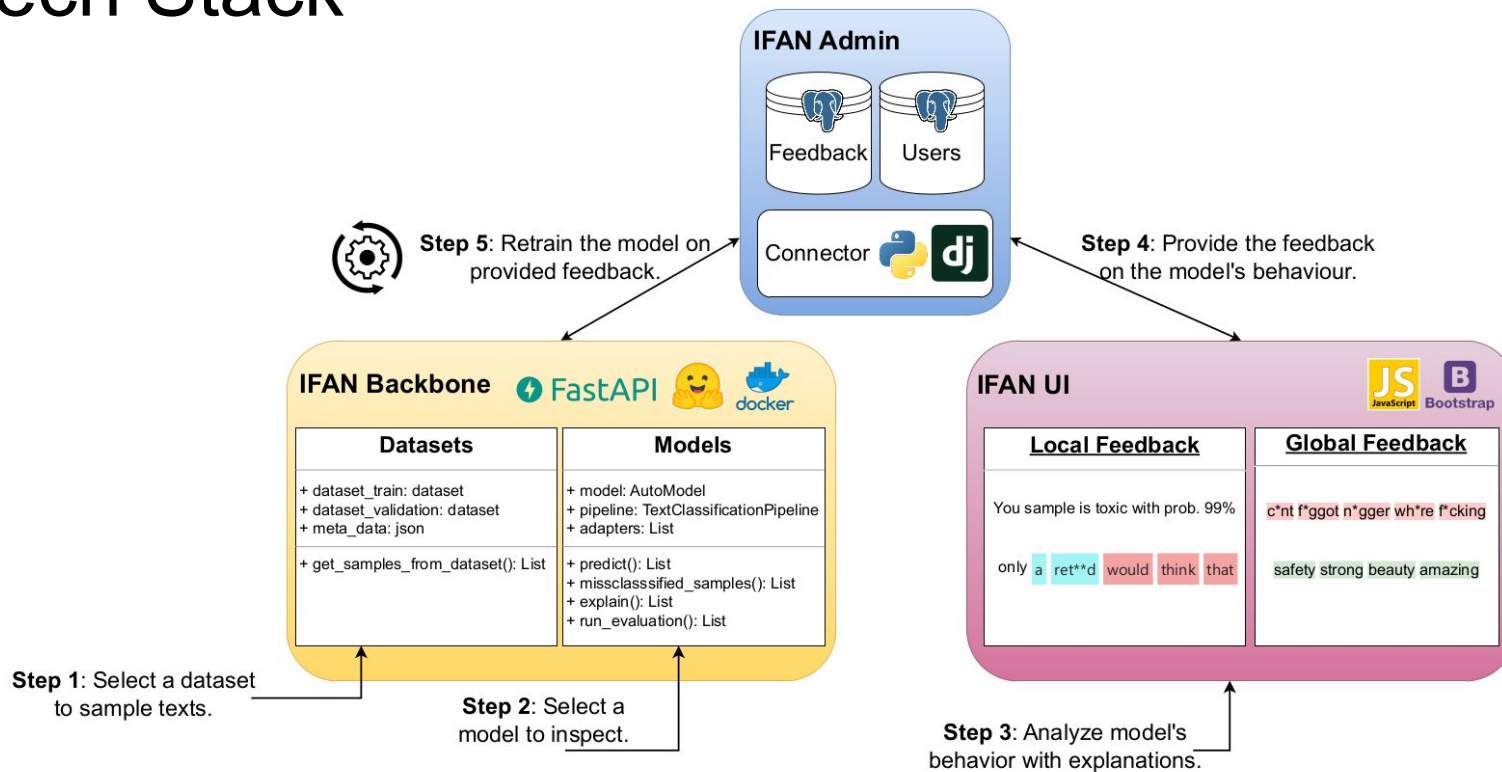
The Idea



The Final Prototype



Tech Stack

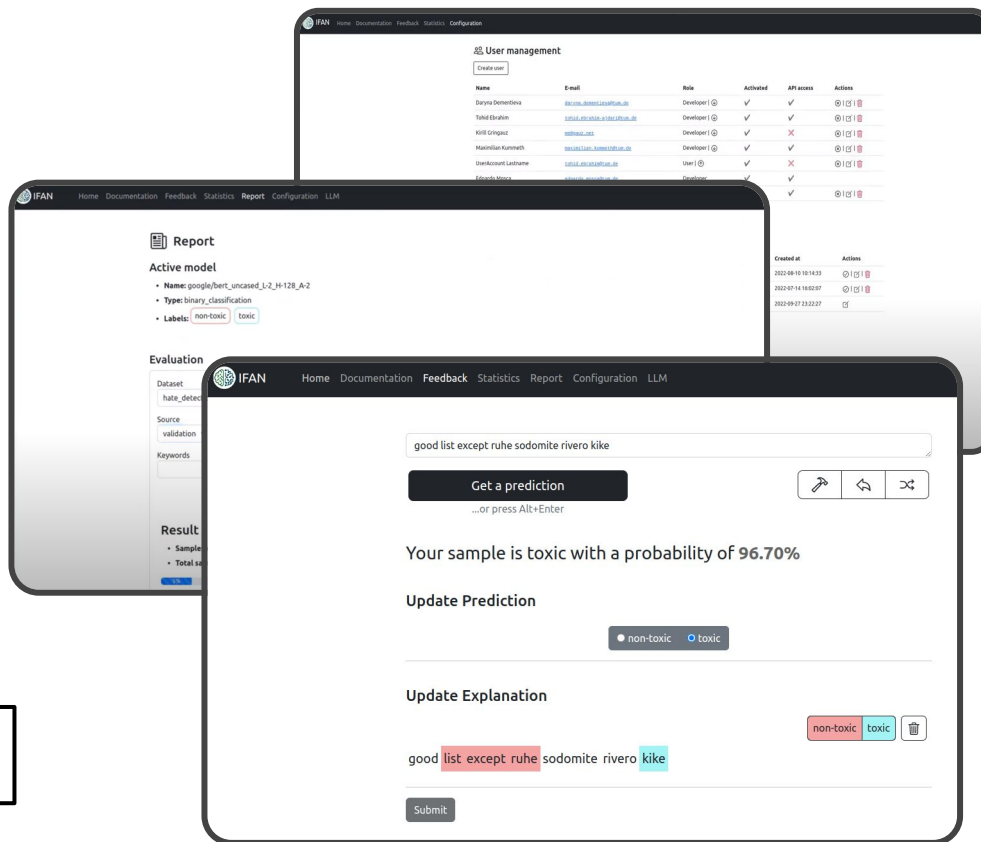


METHODOLOGY

Interface

- Home
- Documentation
- Feedback
- Report
- Configuration

[To the Demo Video](#)



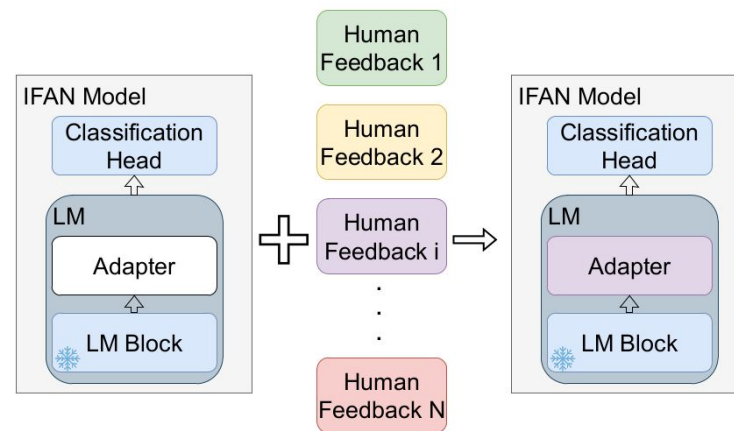
Backbone & API



models ^		
GET	/models/	Get All Models Api
GET	/models/meta_data	Get Meta Data Of Model Api
PATCH	/models/meta_data	Patch Meta Data Of Model Api
GET	/models/misclassified_samples	Get Misclassified Samples Api
GET	/models/evaluation	Get Evaluation Api
GET	/models/available_adapters	Get Available Adapters Api
POST	/models/train_adapter	Post Update Adapter Api
POST	/models/upload	Upload
datasets ^		
GET	/datasets/	Get All Datasets Api
GET	/datasets/meta_data	Get Meta Data Of Dataset Api
GET	/datasets/samples	Get Samples Api
GET	/datasets/samples_with_keyword	Get Samples Keyword Api
POST	/datasets/upload	Upload
prediction ^		
POST	/prediction/	Get Prediction Api
explanation ^		
POST	/explanation/local/	Post Local Explanation Api
POST	/explanation/local/time	Post Local Lime Explanation Api
POST	/explanation/local/captum	Post Local Captum Explanation Api

User Roles & Feedback Mechanism

	Dev	Annotator	Unauthorized
Classification & Explanations	✓	✓	✓
Smart Samples Selection	✓	✓	✗
Feedback	✓	✓	✗
Active Configuration	✓	✗	✗
Model Report & Misl. Samples	✓	✗	✗
New Models & Datasets Upload	✓	✗	✗
New Users Creation	✓	✗	✗



Adapters are trained with batches of human-highlighted sample segments.

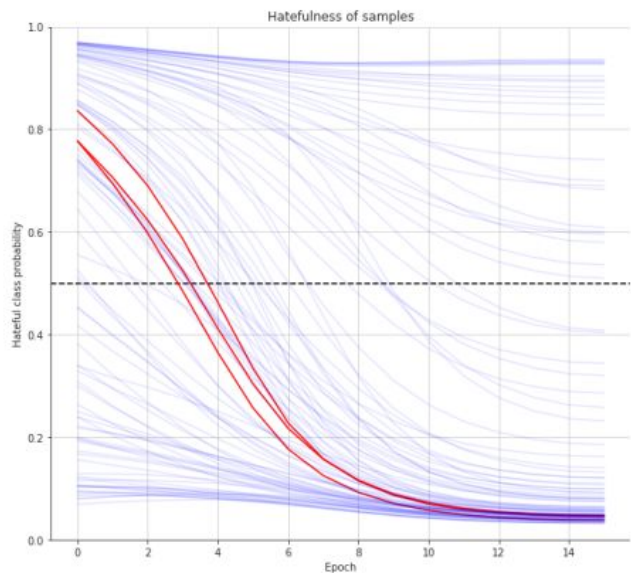
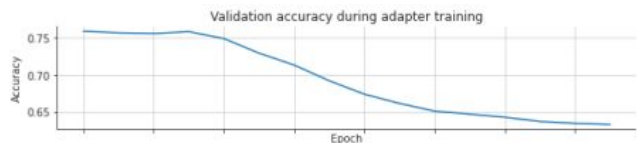
Models love to Forget

- Adapters learn very quickly from feedback (especially on strong signals).
- Models tend to forget what they know, with large drops in performance.
- Adding original samples to feedback batches mitigates the problem (**Rebalancing**).

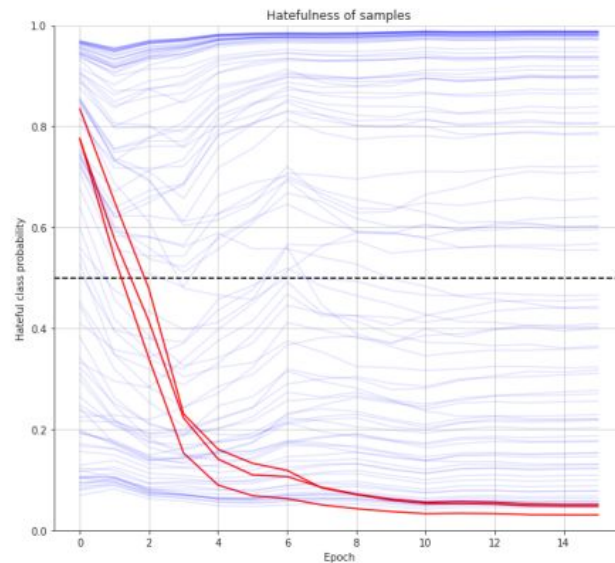
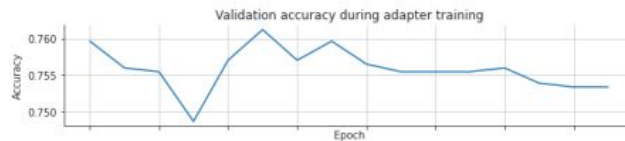
Performance on Use Case

Model	Pr	Re	F1	Pr _J
BERT (baseline)	0.80	0.78	0.79	0.95
<i>Most Confident Missclassified</i>				
BERT+Feedback (non-bal.)	0.34	0.28	0.31	0.82
BERT+Feedback (bal.)	0.78	0.80	0.79	0.97
<i>Least Confident Missclassified</i>				
BERT+Feedback (non-bal.)	0.83	0.73	0.78	0.96
BERT+Feedback (bal.)	0.79	0.78	0.78	0.96

Effects of Rebalancing



Without Rebalancing

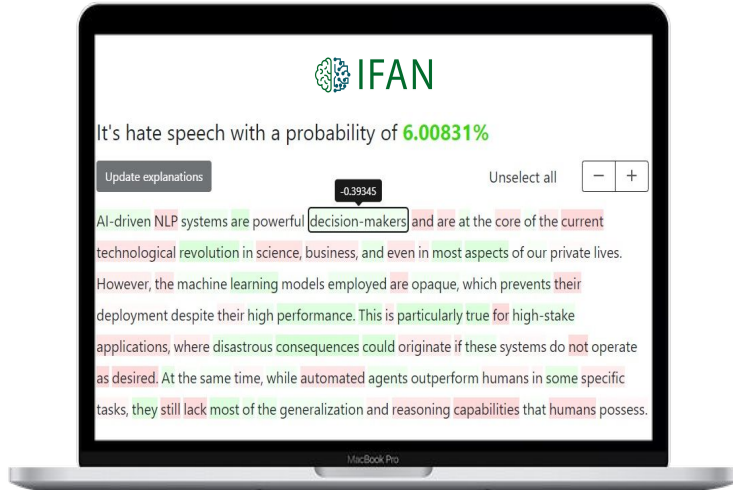


With Rebalancing

Limitations & Takeaways

- Editing explanations can carry human feedback which is beneficial for fixing NLP models.
- As of now we support sentence-to-class and token-to-class tasks (other tasks are work in progress)
- No clear optimal choices of hyperparameters for feedback
- Dealing with a large spectrum of models is hard, but is becoming easier.
- In most cases, your model won't get much better in performance, but is more aligned with humans where it received feedback.
- Rebalancing allows integrating feedback with minimal performance loss.

Thank you!!



Paper



Demo Video

Thank you!!

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